Fizika

9 sinf

MAVZU: Molekulalarning massasi va o‘lchami.

Molekulalar

Moddaning kimyoviy xossasini o‘zida saqlab qoladigan eng kichik zarrasiga molekula deyiladi.

Vodorod molekulasi



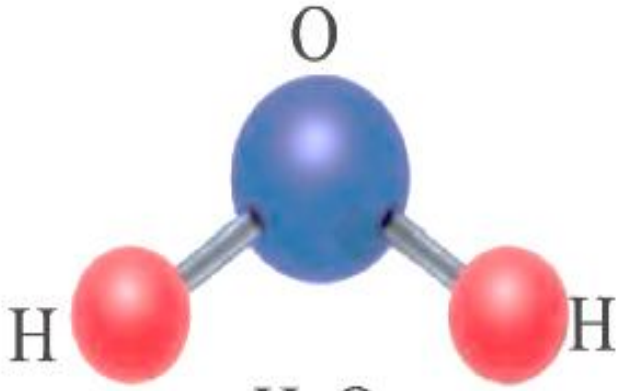
H_2

Kislorod molekulasi



O_2

Suv molekulasi



H_2O

Molekulalarning o‘lchamlari

Molekulaning o‘lchamini aniqlash imkonini beradigan ko‘pgina usullar mavjud. Shunday tajribalardan biri zaytun moyi yordamida o‘tkazilgan.

V= S·d

$$\mathrm{d} = \frac{V}{S} = \frac{1mm^3}{0{,}6m^3} = \frac{10^{-9}m^3}{0{,}6m^2} \approx 1{,}7 \cdot 10^{-9}\mathrm{m}$$

d ≈ 2,3· 10^{-10}m – vodorod molekulasining ,

d≈3·10^{-10}m – suv molekulasining diametrlari.

Molekulalarning massalari

Suv molekulasining diametri $\mathrm{d} \approx 3 \cdot 10^{-10}\, m$

bo‘lsa, hajmi $\mathrm{V} \approx d^3 = (3 \cdot 10^{-10}\mathrm{m})^3$

$$\mathrm{N} = \frac{1\, m^3}{(3 \cdot 10^{-10} m)^3} \approx 3{,}7 \cdot 10^{28} \text{ ta}$$

$m_0 = \frac{1000 kg}{3{,}7 \cdot 10^{28}} \approx 2{,}7 \cdot 10^{-26}\mathrm{kg}$ suv molekulasining massasi.

Nisbiy atom (molekulyar) massa

Atomning massasini ifodalash uchun maxsus atom birligi unit $(u)^2$ kiritilgan.

$$m_{oc} \cdot \frac{1}{12} = 1{,}992 \cdot 10^{-26} kg \cdot \frac{1}{12} = 1{,}66 \cdot 10^{-27} \text{kg} = 1\text{u}$$

$$A_n = \frac{m_o}{\frac{1}{12} m_{oc}} \quad ; \quad A_n = \frac{2{,}66 \cdot 10^{-26} kg}{1{,}66 \cdot 10^{-27} kg} = 16\text{u}$$

Nisbiy atom massasi

Berilgan modda atomi massasining (m_o) uglerod atom massasi (m_{oc}) 1/12 qismining nisbatiga, shu moddaning nisbiy atom massasi deyiladi.

1- mashq 1-masala

Massasi 2,4 kg bo‘lgan ko‘mirda qancha uglerod atomi borligini hisoblang. Uglerod atomining massasini $2 \cdot 10^{-26}$ kg ga teng deb oling.

Berilgan:

$m = 2{,}4\ kg$

$m_0 = 2 \cdot 10^{-26} kg$

Topish kerak: $N - ?$

Formula:

$$N = \frac{m}{m_0}$$

Yechilishi:

$$N = \frac{2{,}4\ kg}{2 \cdot 10^{-26} kg} = 1{,}2 \cdot 10^{26}$$

Javob: $N = 1{,}2 \cdot 10^{26}\ ta$

1-mashq 4-masala

Idishdagi suvda $10^{24}\ ta$ suv molekulasi bo‘lsa, suvning hajmi qanday? Suv molekulasining diametri $\mathrm{d} = 3 \cdot 10^{-10}\ m$ ga teng deb oling.

Berilgan:

$N = 10^{24} ta$

$d = 3 \cdot 10^{-10} m$

Topish kerak: $V - ?$

Formula:

$N = \frac{V}{V_0}$

$V = V_0 \cdot N, \quad V_0 = \frac{4}{3}\pi R^3, \quad R = \frac{d}{2}$

$V_0 = \frac{4}{3}\pi(\frac{d}{2})^3 \Rightarrow V = N \cdot \frac{4}{3}\pi(\frac{d}{2})^3$

Yechish:

$$V = 10^{24} \cdot \frac{4}{3} \cdot 3{,}14 \cdot \left(\frac{3 \cdot 10^{-10} m}{2}\right)^3 = 14{,}13 \cdot 10^{-6} m^3$$

$$= 14{,}13\ sm^3 = 14{,}13\ ml$$

Javob: $\boldsymbol{V = 14,13\ ml}$

Mustaqil bajarish uchun topshiriqlar

1. Mavzuga doir savollarga javob yozish. (11-sahifa).
2. 1-mashq, 2- va 3-masalalarni yechish.